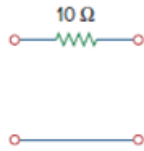


## Problem

For the single-element two-port network in Fig. 19.64(b), **B** is:

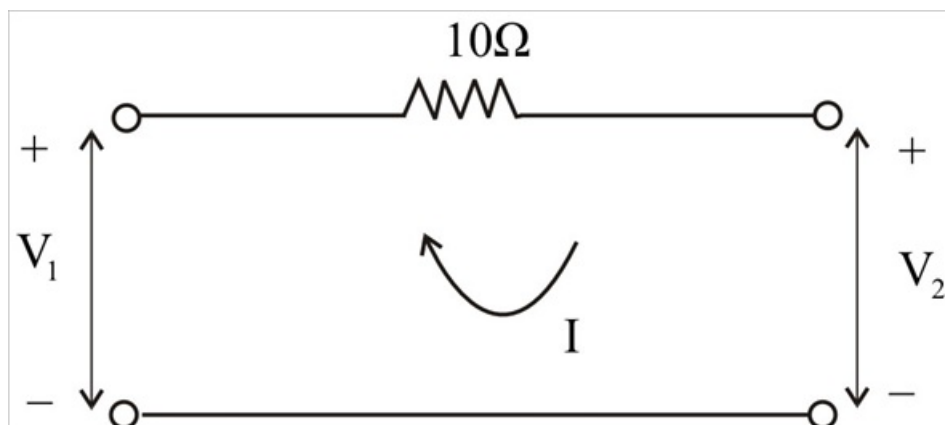
- (a) 0
- (b) 5
- (c) 10
- (d) 20
- (e) undefined

**Figure 19.64(b)**



## Step-by-step solution

## Step 1 of 2



## Step 2 of 2

We have  $10I = V_1 - V_2$  ( $|I_1| = |I_2| = I$ )

Now,

$$B = -\frac{V_1}{I_2} \bigg|_{V_2=0}$$

$$\Rightarrow 10I_2 = V_1$$

$$\text{pr. } B = -\frac{V_1}{I_2} = -10$$